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Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

A CAPSTONE PROJECT REPORT

CLOUD-BASED STUDENT AND TEACHER FEEDBACK MANAGEMENT SYSTEM WITH REAL-TIME ANALYTICS

Submitted in partial fulfilment of the requirements
for the course

CLOUD COMPUTING

for the award of the degree of
BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY

Submitted by

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Under the Guidance of

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “CLOUD-BASED STUDENT AND TEACHER FEEDBACK MANAGEMENT SYSTEM WITH REAL-TIME ANALYTICS” has been carried out by A.VINAY(192521441) and under the supervision of Dr.K.Santhiya Lakshmi and is submitted in partial fulfilment of the requirements for the current semester of the B.tech ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, A.VINAY(192521441) of the B.TECH INFORMATION TECHNOLOGY, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “CLOUD-BASED STUDENT AND TEACHER FEEDBACK MANAGEMENT SYSTEM WITH REAL-TIME ANALYTICS” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

INDIVIDUAL CONTRIBUTION STATEMENT

This is an individual report for a team capstone. The team presentation identifies four students and four functional modules, but it does not explicitly assign individual responsibilities. Therefore, the table below records the team structure while keeping the individual contribution wording conservative. The exact allocation should be aligned with the team's actual work log or repository before final submission.

Student / Register No.	Project Role / Module Area	Indicative Contribution	Share
Dharshan T / 192421100	Team member	Module 1 – User Authentication & Role Management: Requirement analysis, login/registration, role-based access, user management, documentation and testing.	25%
Sanjay P S / 192525429	Individual report owner	Module 2 – Feedback Collection & Management: Feedback forms, ratings/comments, anonymous feedback, validation and duplicate-submission prevention.	25%
Vinay A / 192521441	Team member	Module 3 – Analytics & Report Generation: Rating calculations, dashboards, charts, subject/department analysis and PDF/Excel reports.	25%
Pranava Sorupan B / 192511277	Team member	Module 4 – Cloud Data Storage: Firebase Firestore integration, cloud storage, real-time synchronization, data management and integration support.	25%

ABSTRACT

Educational institutions require an efficient system to collect feedback from students about teaching quality and courses. Traditional feedback methods are often paper-based, time-consuming, and difficult to manage. They may require considerable administrative effort and can make it difficult to analyze feedback quickly.

The **Cloud-Based Student and Teacher Feedback Management System with Real-Time Analytics** provides a digital solution for collecting, storing, analyzing, and presenting student feedback. Cloud computing enables secure, scalable, and real-time storage and access to feedback information. Students can submit feedback online, while teachers and administrators can access analytical reports and dashboards.

The system consists of four major modules: **User Authentication & Role Management, Feedback Collection & Management, Analytics & Report Generation, and Cloud Data Storage.**

My individual contribution is **Module 3 – Analytics & Report Generation**. This module converts collected feedback into meaningful information. It calculates average ratings, provides subject-wise and department-wise analysis, displays feedback trends through charts, compares teacher performance across semesters, and generates downloadable PDF or Excel reports.

The analytics module helps teachers and administrators understand feedback results in a simple and visual form. Dashboards and charts make it easier to identify performance trends and compare feedback results. Reports provide a structured way to store and communicate analytical findings.

The project demonstrates how cloud computing can be applied to an educational environment to improve feedback management, accessibility, transparency, and decision-making.

Keywords: Cloud Computing, Student Feedback, Teacher Evaluation, Real-Time Analytics, Dashboard, Data Visualization, Firebase Firestore, Report Generation.

TABLE OF CONTENTS

Sl. No.	Title	Page No.
i	Certificate from the Supervisor	ii
ii	Declaration by the Candidate	iii
iii	Individual Contribution Statement	iv
iv	Abstract	v
v	List of Figures	vi
vi	List of Tables	vii
vii	List of Abbreviations / Symbols	viii
1	Chapter 1: Introduction and Engineering Problem	1
2	Chapter 2: Literature Review and Existing Solutions	5
3	Chapter 3: Engineering Design, Trade-offs, Sustainability, Ethics, Safety, Risk & Security	9
4	Chapter 4: Implementation, Testing & Results, Project Management	15
5	Chapter 5: Conclusion, Future Work and Reflection	22
	References	26
	Appendices	27

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 3.1	Proposed cloud-based system architecture	10
Figure 4.1	FeedbackCloud login interface	16
Figure 4.2	Administrator dashboard	16
Figure 4.3	Teacher dashboard	17
Figure 4.4	Student dashboard	17
Figure 4.5	Cloud data storage and synchronization concept	18

LIST OF TABLES

Table No.	Table Name	Page No.
Table 1.1	Project objectives	3
Table 2.1	Summary of reviewed literature	6
Table 2.2	Comparison of existing and proposed approach	7
Table 3.1	Stakeholder and user requirements	9
Table 3.2	Functional and non-functional requirements	10
Table 3.3	Module design specifications	11
Table 3.4	Alternative solution evaluation	12
Table 3.5	Risk register	13
Table 4.1	Development resources and tools	15
Table 4.2	Test plan and verification	19
Table 4.3	Objective achievement	20
Table 4.4	Project milestones	21

LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation	Description	Meaning / Use
API	Application Programming Interface	Interface for communication between software components
CRUD	Create, Read, Update, Delete	Common data-management operations
DB	Database	Structured storage of application data
UI	User Interface	Human-computer interaction layer
RBAC	Role-Based Access Control	Access control according to user role
SaaS	Software as a Service	Cloud-delivered application model
Firebase	Google Firebase platform	Cloud services used for authentication/data services
Firestore	Cloud Firestore	Cloud-hosted NoSQL database
PDF	Portable Document Format	Downloadable report format
Excel	Microsoft Excel spreadsheet format	Downloadable tabular report format

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Educational institutions require an efficient method to collect student feedback regarding teaching quality and courses. Feedback provides useful information for understanding students' opinions and identifying areas that require improvement.

Traditional feedback systems generally depend on paper forms. Such systems require manual distribution, collection, storage and analysis. This makes the feedback process slow and difficult to manage.

Cloud computing provides an opportunity to improve this process by allowing information to be stored and accessed through cloud-based services. The proposed system provides online feedback submission and allows teachers and administrators to view analytical reports.

My contribution focuses on the **analytics stage**, where collected feedback is converted into useful information through calculations, dashboards, charts and reports.

1.2 Need for the Project

The major problems with the traditional feedback process are:

- Manual feedback collection is slow.
- Paper forms require physical storage.
- Records may be lost or duplicated.
- Manual analysis requires additional effort.
- Feedback results may be available only after considerable delay.
- Lack of anonymity may affect student responses.
- Traditional systems have limited accessibility.
- Real-time analytical information is generally unavailable.

1.3 Problem Statement

Manual feedback collection is time-consuming and requires significant administrative effort. Paper-based forms can result in data loss, duplication and inaccurate record keeping.

Feedback analysis is also difficult because responses must be manually organized and calculated. Students may hesitate to provide honest opinions when anonymity is not available.

Therefore, there is a need for a cloud-based system that provides centralized feedback management together with real-time analytics.

1.4 Project Aim

The aim of this project is to develop a **secure, scalable and efficient cloud-based feedback management system** that enables students to submit feedback and allows teachers and administrators to analyze feedback through dashboards and reports.

1.5 Project Objectives

The project objectives are:

1. To develop a secure cloud-based platform for collecting student feedback.
2. To provide role-based access for Admin, Teachers and Students.
3. To store feedback securely in a cloud database.
4. To provide real-time synchronization.
5. To generate analytical reports and dashboards.
6. To improve the efficiency, transparency and accessibility of the feedback process.

No.	Objective
1	To develop a secure cloud-based platform for collecting student feedback on teachers and courses.
2	To provide role-based access for Admin, Teachers, and Students using cloud authentication.
3	To store feedback securely in a cloud database with real-time synchronization.
4	To generate analytical reports and dashboards for evaluating teaching performance.
5	To improve the efficiency, transparency, and accessibility of the feedback process.

1.6 Scope

The project covers:

- Student feedback collection.
- Teacher and course evaluation.
- User authentication.
- Role-based access.
- Cloud data storage.
- Real-time synchronization.
- Feedback analytics.
- Dashboard visualization.
- Subject-wise analysis.

- Department-wise analysis.
- Semester-based performance comparison.
- PDF/Excel report generation.

1.7 Expected Engineering Outcome

The expected outcome is a cloud-based feedback platform that reduces manual work and provides timely analytical information.

The analytics component provides a way for teachers and administrators to understand feedback through ratings, charts, trends and reports.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted to understand existing cloud-based feedback systems and academic evaluation platforms.

The reviewed studies focus on cloud computing, Firebase, real-time feedback analysis, cloud databases, authentication, dashboards and data visualization.

2.2 Summary of Reviewed Literature

Year	Author(s)	Study / Focus	Methodology / Technology	Key Finding	Limitation
2025	Sharma et al.	Cloud-Based Feedback Management System for Higher Education	Cloud Computing, Firebase, RBAC	Improved feedback collection and secure real-time access.	No AI sentiment or predictive analytics.
2024	Kumar & Singh	Real-Time Student Feedback Analysis Using Cloud Technologies	Firestore, Chart.js, visualization	Enabled instant analytical reports.	Limited high-load scalability evaluation.
2023	Patel et al.	Secure Online Feedback System Using Cloud Database	Cloud database, authentication, encryption	Improved security and accessibility.	Anonymous feedback and advanced reporting not supported.
2022	Ahmed & Rahman	Web-Based Faculty Evaluation System with Cloud Integration	Web application, cloud storage, dashboard	Centralized faculty evaluation.	Focused mainly on faculty feedback.
2021	Lee et al.	Cloud-Enabled Academic Feedback Management	SaaS architecture, cloud database	Scalable centralized data management.	No real-time visualization/downloadable reports.

2.3 Comparative Analysis

Approach	Advantages	Limitations	Relevance to Proposed Work
Paper-based feedback	Simple and familiar	Slow, difficult to analyze, physical storage	Baseline problem addressed by digital platform
Basic online feedback	Faster collection, easier access	May lack cloud synchronization and analytics	Provides foundation for online forms
Cloud feedback storage	Centralized, scalable, accessible	Requires secure access and data governance	Core storage approach
Dashboard-based analysis	Visual and easier to interpret	Depends on data quality and correct aggregation	Core analytics approach
Proposed integrated system	Combines authentication, feedback, cloud storage, analytics and reports	Requires careful security and role management	Target architecture

2.4 Research Gap

The literature indicates that existing systems provide different combinations of feedback collection, cloud storage and analytics.

However, some systems lack:

- Real-time visualization.
- Downloadable reports.
- Anonymous feedback.
- Course-level analysis.
- Integrated analytical dashboards.

The proposed system combines these functions into a single cloud-based platform.

2.5 Proposed Contribution

The proposed system provides an integrated platform for feedback collection, cloud storage and analytical reporting.

My module contributes specifically by providing the **analytics and report generation layer**. It converts feedback records into ratings, trends, comparisons and downloadable reports.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholder and User Requirements

Stakeholder	Requirement
Student	Secure access, easy feedback submission, rating/comment support, optional anonymity, submission confirmation.
Teacher	Secure access to feedback summaries, performance trends, and reports relevant to teaching.
Administrator	Manage users and feedback structures; access institution-level analytics and reports.
Institution	Centralized, reliable, scalable feedback data with timely analytical information.

3.2 Functional and Non-Functional Requirements

Requirement	Type	Measurable / Expected Behaviour
User registration/login	Functional	Valid users can authenticate and enter the correct role dashboard.
Role-based access	Functional	Students, teachers and administrators see role-appropriate functions.
Feedback submission	Functional	Students can submit ratings/comments through an online form.
Duplicate prevention	Functional	The same feedback session cannot be submitted repeatedly.
Cloud storage	Functional	Feedback records are stored in the cloud database.
Real-time synchronization	Functional	Updated data is available across authorized views.
Analytics	Functional	Ratings and feedback are summarized through charts/reports.

Security	Non-functional	Authentication and controlled access protect system data.
Scalability	Non-functional	Cloud storage can accommodate increasing feedback records.
Usability	Non-functional	Core workflows are understandable to intended users.

3.3 Proposed System Architecture

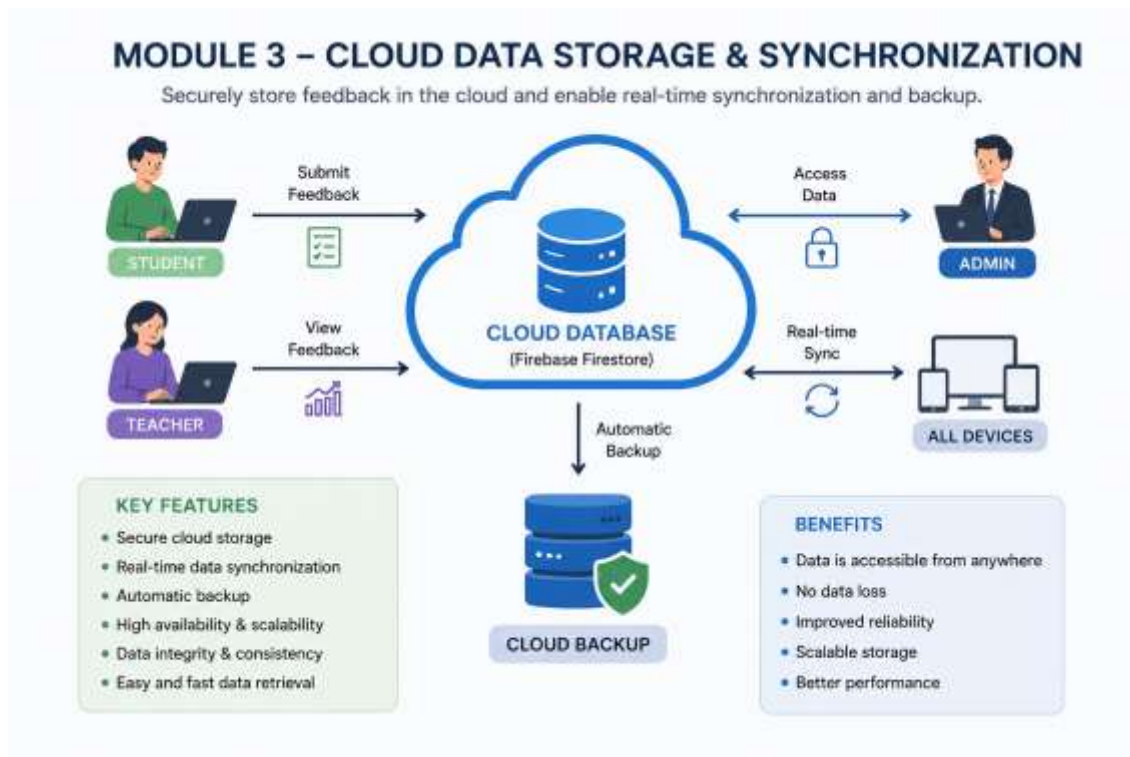


Figure 3.1. Cloud data storage and synchronization concept represented in the team presentation.

The architecture is organized around four functional areas. Authentication controls entry into the system; feedback management handles collection and validation; cloud data storage provides centralized persistence and synchronization; and analytics/report generation transforms stored feedback into useful summaries. The team presentation explicitly identifies Firebase Firestore for centralized data management and real-time synchronization.

3.4 Module Design Specifications

Module	Main Functions	Cloud / Technical Focus
Module 1: User Authentication & Role Management	Secure login/registration; Student, Teacher and Admin roles; role-based access; profile management; password recovery.	Cloud authentication and access control.
Module 2: Feedback Collection & Management	Online forms; teacher/course ratings; comments; anonymous feedback option; duplicate-submission prevention.	Cloud-connected forms and database operations.
Module 3: Analytics & Report Generation	Average ratings; subject/department analysis; trends; semester comparison; PDF/Excel reports.	Cloud data retrieval, aggregation and visualization.
Module 4: Cloud Data Storage	Secure storage; timestamped records; real-time synchronization; scalability; backup/accessibility.	Firebase Firestore and cloud services.

3.5 Alternative Solutions and Trade-offs

Alternative	Benefit	Limitation	Decision
Paper-based system	Low technical setup	Manual, slow, hard to analyze	Rejected
Standalone local database	Simple local control	Limited accessibility/scalability	Rejected
Basic web feedback system	Easy online collection	Limited cloud-native capabilities	Partial baseline
Cloud-based integrated system	Accessible, scalable, centralized, real-time analytics	Requires cloud security and configuration	Selected

3.6 Security Considerations

The project considers secure authentication and controlled access to feedback information.

Analytics should only display information to authorized teachers and administrators.

Sensitive student information should not be unnecessarily exposed through analytical dashboards.

3.7 Sustainability and Ethics

The cloud-based system reduces dependence on paper-based records.

Digital storage reduces physical documentation and makes feedback information easier to manage.

Analytics should be presented fairly and accurately. Incorrect calculations or misleading visualizations could result in incorrect decisions.

Student privacy should also be protected when feedback is analyzed and reported

3.8 Risk Register

Risk	Likelihood	Impact	Mitigation
Unauthorized access to feedback	Medium	High	Authentication, RBAC, secure database rules.
Duplicate feedback submissions	Medium	Medium	Validate user/session before accepting a submission.
Incorrect analytics aggregation	Medium	High	Validate records, timestamps and calculation logic.
Cloud/service unavailability	Low–Medium	High	Use reliable cloud services and backup/recovery procedures.
Loss of anonymity	Low–Medium	High	Separate identity information from anonymous feedback

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Resources and Tools

Resource / Technology	Purpose in the Proposed System
Firebase Authentication	Secure authentication for system users.
Firebase Firestore	Centralized cloud storage for feedback and user-related records.
Cloud services	Real-time access, synchronization, scalability and reliable data management.
Web interface	Provides dashboards and feedback forms for different user roles.
Chart-based visualization	Displays ratings, trends and performance summaries.
PDF / Excel reporting	Provides downloadable summarized feedback reports.

4.2 Implementation Evidence

The implemented system provides different interfaces for students, teachers and administrators.

Students submit feedback through online forms. Feedback is stored in the cloud. The analytics module processes the stored feedback and generates useful information.

The main implementation flow is:

Login → Feedback Submission → Cloud Storage → Data Processing → Analytics → Dashboard → Report

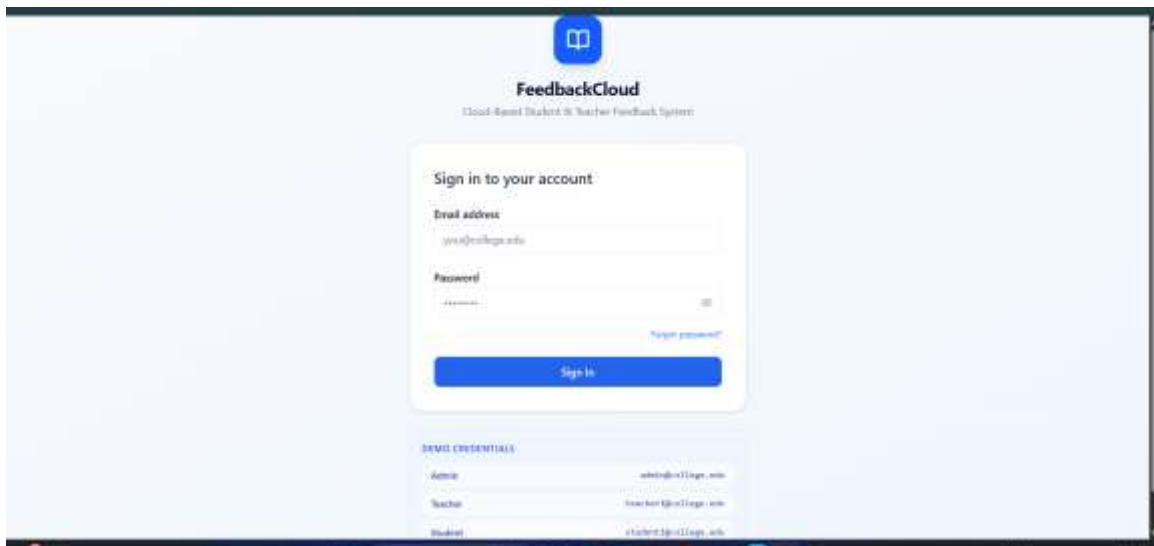


Figure 4.1. FeedbackCloud login interface shown in the team implementation slide.

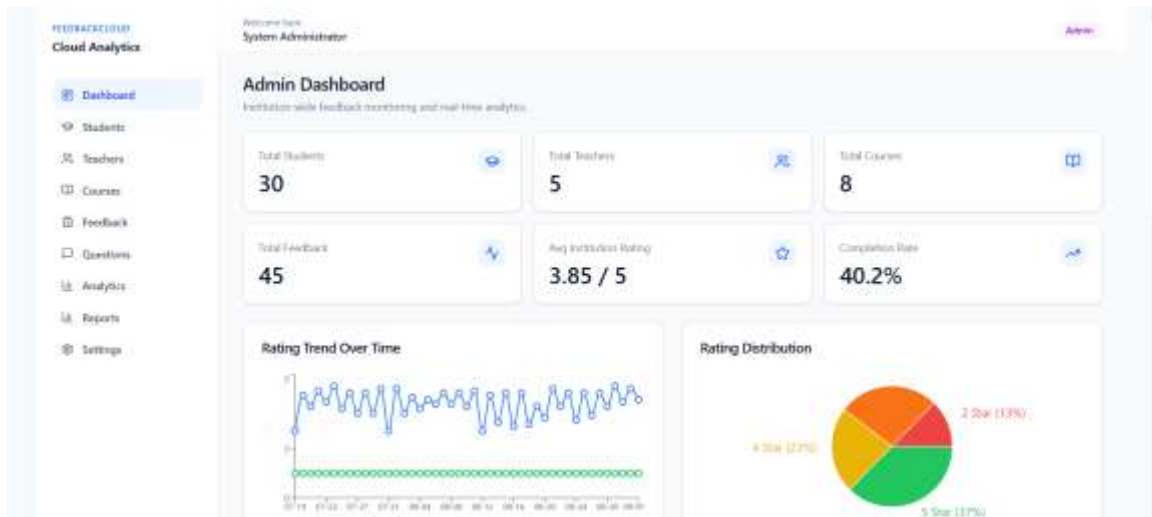


Figure 4.2. Administrator dashboard showing user and feedback analytics.

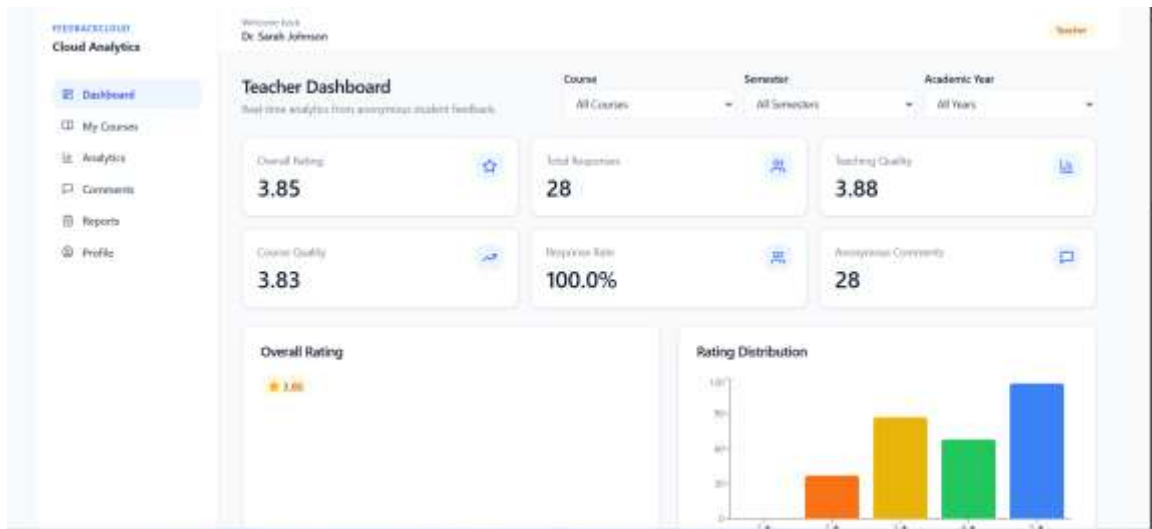


Figure 4.3. Teacher dashboard with course and feedback performance information.

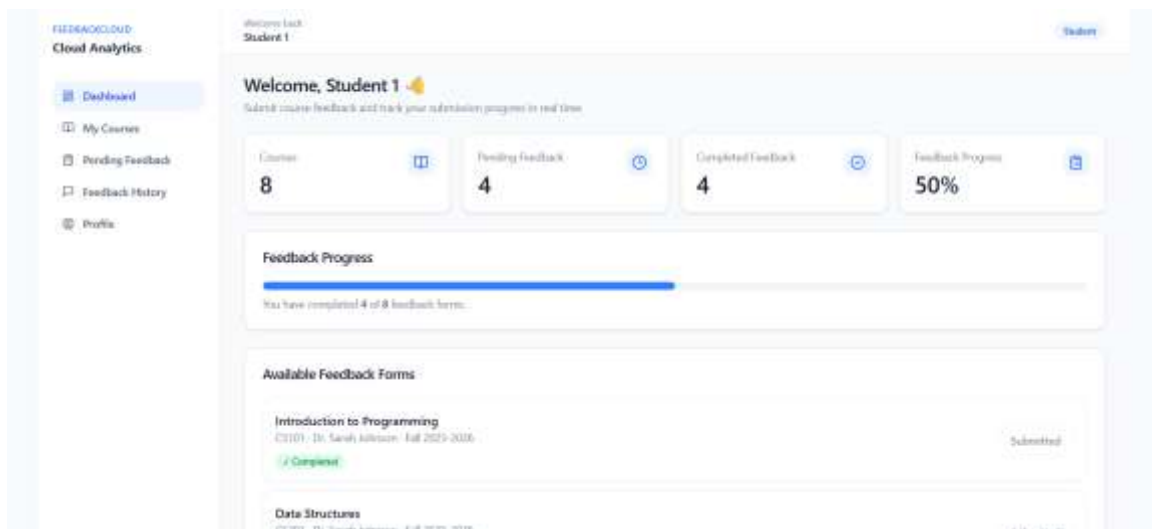


Figure 4.4. Student dashboard showing available feedback forms and progress.

4.3 Functional Workflow

Step 1 – User Login

The user enters the system using the appropriate credentials.

Step 2 – Feedback Collection

Students select a teacher/course and submit ratings and comments.

Step 3 – Cloud Storage

The feedback is stored in the cloud database.

Step 4 – Data Processing

Feedback records are retrieved for analysis.

Step 5 – Rating Calculation

Average ratings are calculated for teachers and courses.

Step 6 – Analytics

Feedback is grouped and analyzed according to subjects, departments and semesters.

Step 7 – Dashboard

The results are displayed using charts and dashboard elements.

Step 8 – Report Generation

Summarized results are generated as PDF or Excel reports.

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4.4 Test Plan and Verification

Requirement	Test Method	Acceptance Criterion
Login/authentication	Use valid and invalid credentials.	Valid users authenticate; invalid credentials are rejected.
Role-based access	Login as Student, Teacher and Admin.	Each role receives only its intended functions.
Feedback submission	Submit a complete feedback form.	Feedback is accepted and stored correctly.
Anonymous feedback	Submit feedback using anonymous option.	Identity is not exposed in the feedback view.
Duplicate prevention	Attempt a second submission for the same session.	Duplicate submission is blocked.
Cloud storage	Create and retrieve feedback records.	Records are available from the cloud database.
Real-time synchronization	Update data and refresh authorized views.	Updated information becomes available across supported views.
Analytics	Create sample feedback and inspect dashboard.	Aggregated ratings and trends are displayed correctly.
Report generation	Generate PDF/Excel report.	Report contains the selected summarized feedback information.

4.5 Expected / Demonstrated Outcomes

Objective	Evidence / Outcome
Secure cloud platform	Authentication and role-based access are specified in Module 1.
Role-based access	Separate Student, Teacher and Administrator roles are defined.
Cloud storage	Firebase Firestore is specified for centralized data management.
Real-time synchronization	Cloud data synchronization across users/devices is specified.
Analytics and reports	Dashboards, charts and PDF/Excel reports are specified in Module 3.
Improved accessibility	Cloud-based access allows feedback services to be used through the web application.

4.6 Project Milestones

Milestone	Planned Activity	Completion Evidence
M1	Problem definition and literature review	Problem statement, objectives and reviewed studies.

M2	Authentication and feedback module foundation	Login, role structure and feedback workflow.
M3	Cloud data storage	Firestore-based centralized storage and synchronization design.
M4	Analytics and reporting	Dashboards, charts and downloadable report workflow.
M5	Integration	Connected user, feedback, storage and analytics modules.
M6	Testing and documentation	Functional verification, screenshots and individual report.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **Cloud-Based Student and Teacher Feedback Management System with Real-Time Analytics** provides a modern solution for managing academic feedback.

The project addresses the limitations of traditional paper-based feedback by providing online feedback collection, cloud-based storage and analytical reporting.

The cloud architecture provides scalable and accessible storage, while the analytics dashboard helps teachers and administrators make informed decisions based on feedback.

My contribution through **Module 3 – Analytics & Report Generation** helped transform feedback data into useful information through rating calculations, dashboards, charts, subject-wise analysis, department-wise analysis and downloadable reports.

5.2 Future Work

The system can be improved by adding:

1. **AI-based sentiment analysis** for student comments.
2. **Predictive analytics** for identifying future feedback trends.
3. Mobile application support.
4. Advanced dashboard customization.
5. More visualization techniques.
6. Automated email notifications.
7. Advanced report customization.
8. Larger-scale performance testing.
9. Improved security monitoring.
10. Automated cloud backup.

The team literature review also identifies AI-based sentiment analysis and predictive analytics as possible future enhancements.

5.3 Professional Learning

5.4 Individual Reflection – Dharshan T (192421100)

Working on this project gave me practical experience in developing an analytics-based component of a cloud computing application. My primary responsibility was **Module 3 – Analytics & Report Generation**, which helped me understand how raw feedback data can be converted into meaningful information for teachers and administrators.

One of the important things I learned was how to calculate average ratings from student feedback. I understood that accurate calculations are essential because the results displayed on dashboards and reports depend directly on the quality of the processed data.

I also learned how dashboards and charts can make complex feedback information easier to understand. Instead of presenting only raw data, visualizations can show trends, comparisons and performance information in a simple manner. This helped me understand the importance of data visualization in decision-making.

Another important part of my contribution was subject-wise and department-wise analysis. I learned how feedback can be grouped into different categories to identify performance patterns. Semester comparison also helped me understand how feedback can be used to observe changes in teacher performance over time.

Working on PDF and Excel report generation improved my understanding of how analytical results can be converted into useful documents. Reports provide a convenient way for administrators and teachers to save, review and share feedback results.

Testing was another valuable part of my learning. I understood that analytical modules must be tested carefully because even a small calculation or data-processing error can affect the final dashboard and report. I learned to check whether the calculated results matched the expected values.

Overall, this project improved my technical knowledge of **cloud computing, analytics, dashboards, data visualization and report generation**. It also improved my teamwork, communication, problem-solving and documentation skills. My contribution to Module 3 helped me understand how cloud-based feedback data can be transformed into useful information for academic decision-making.

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APPENDICES

Appendix A – Module Summary

Module	Task	Specific Parameters	Outcome / Outcome Parameters	Techniques / Technologies
Module 1 – User Authentication & Role Management	Implement secure access and user-role management.	Credentials, roles, profiles, password recovery.	Secure authentication and controlled access.	Cloud authentication, RBAC, web interface.
Module 2 – Feedback Collection & Management	Collect and manage student feedback.	Teacher/course, ratings, comments, anonymity, submission status.	Efficient and organized feedback collection.	Online forms, validation, cloud database operations.
Module 3 – Analytics & Report Generation	Analyze feedback and generate reports.	Average ratings, trends, subject/department analysis, semester comparison.	Actionable insights and downloadable reports.	Data aggregation, charts, PDF/Excel reporting.
Module 4 – Cloud Data Storage	Store and synchronize feedback data in the cloud.	Records, timestamps, synchronization, scalability, backup.	Reliable, centralized and accessible data.	Firebase Firestore, cloud services, real-time synchronization.

Appendix B – Evidence Included from the Team Presentation

- Login interface showing the entry point for authenticated users.
- Administrator dashboard showing system-level analytics.
- Teacher dashboard showing course and feedback performance information.
- Student dashboard showing feedback forms and progress.
- Cloud data storage and synchronization concept diagram.

Appendix C – Finalization Checklist

- Confirm the exact individual contribution with the team work log.
- Confirm the final cloud services actually used in the implemented system.
- Insert the final course code if required by the department.
- Verify references and add DOI/URL details if the university requires them.
- Update page numbers in the Table of Contents after opening the document in Microsoft Word.